

Ziyi Chen, Ph.D.



<https://scholar.google.com/citations?user=zjSBVOIAAAAJ&hl=en&oi=ao>

Research Interest

- Reinforcement Learning** Computation complexity, distributed learning, model robustness, safety
- Optimization for Large Scale Machine Learning** Nonconvex optimization, stochastic optimization, federated/decentralized optimization, minimax optimization, bilevel optimization
- Large Language Models** Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO)

Academic Appointments

- 2023 – present **Postdoctoral Research Associate, University of Maryland (College Park)**
Department of Computer Science
Advisor: Prof. Heng Huang

Education

- 2019 – 2023 **Ph.D. Electrical & Computer Engineering, University of Utah**
Thesis title: *Convergence Analysis of Minimax Optimization and Multiagent Reinforcement Learning.*
Advisor: Prof. Yi Zhou
- 2015 – 2019 **M.S. Statistical Science, Cornell University**
- 2012 – 2015 **M.S. Applied Mathematics, Peking University**
- 2008 – 2012 **B.S. Information & Computing Science, Xi'an Jiaotong University.**

Research Publications

Journal Articles and Preprints

- 1** **Z. Chen** and H. Huang, “Achieve performatively optimal policy for performative reinforcement learning,” *ArXiv:2510.04430*, 2025. URL: <https://arxiv.org/pdf/2510.04430>.
- 2** **Z. Chen**, J. Li, P. Yu, and H. Huang, “Provably mitigating corruption, overoptimization, and verbosity simultaneously in offline and online RLHF/DPO alignment,” *ArXiv:2510.05526*, 2025. URL: <https://arxiv.org/pdf/2510.05526>.
- 3** **Z. Chen**, P. Yu, and H. Huang, “Zeroth-order methods for stochastic nonconvex nonsmooth composite optimization,” *ArXiv:2510.04446*, 2025. URL: <https://arxiv.org/pdf/2510.04446>.
- 4** **Z. Chen**, S. Zhang, and H. Huang, “Trade-off in estimating the number of byzantine clients in federated learning,” *ArXiv:2510.04432*, 2025. URL: <https://arxiv.org/pdf/2510.04432>.
- 5** S. Ma, **Z. Chen**, Y. Zhou, and H. Huang, “Rectified robust policy optimization for model-uncertain constrained reinforcement learning without strong duality,” *Transactions on Machine Learning Research*, 2025. URL: <https://openreview.net/forum?id=7l63xwAgAW>.

- 6 **Z. Chen**, Z. Hu, Q. Li, Z. Wang, and Y. Zhou, “A cubic regularization approach for finding local minimax points in nonconvex minimax optimization,” *Transactions on Machine Learning Research (TMLR)*, 2023.  URL: <https://openreview.net/forum?id=jVMdg31De>.
- 7 **Z. Chen**, B. Kailkhura, and Y. Zhou, “An accelerated proximal algorithm for regularized nonconvex and nonsmooth bi-level optimization,” *Machine Learning*, vol. 112, pp. 1433–1463, 2023.  URL: <https://link.springer.com/article/10.1007/s10994-023-06329-6>.
- 8 S. Ma, **Z. Chen**, S. Zou, and Y. Zhou, “Decentralized robust v-learning for solving markov games with model uncertainty,” *Journal of Machine Learning Research (JMLR)*, vol. 24, no. 371, pp. 1–40, 2023.  URL: <https://www.jmlr.org/papers/v24/23-0310.html>.
- 9 **Z. Chen**, Y. Zhou, and R.-R. Chen, “Multi-agent off-policy tdc with near-optimal sample and communication complexities,” *Transactions on Machine Learning Research (TMLR)*, 2022.  URL: [https://openreview.net/forum?id=tnPjQpYk7D&referrer=%5BTMLR%5D\(%2Fgroup%3Fid%3DTMLR\)](https://openreview.net/forum?id=tnPjQpYk7D&referrer=%5BTMLR%5D(%2Fgroup%3Fid%3DTMLR)).

Conference Proceedings

- 1 Q. He, P. Yu, **Z. Chen**, and H. Huang, “Revisiting convergence: A study on shuffling-type gradient methods,” in *International Conference on Machine Learning (ICML)*, 2025.  URL: [https://openreview.net/forum?id=pXoUfvNRRu&referrer=%5Bthe%20profile%20of%20Ziyi%20Chen%5D\(%2Fprofile%3Fid%3D~Ziyi_Chen2\)](https://openreview.net/forum?id=pXoUfvNRRu&referrer=%5Bthe%20profile%20of%20Ziyi%20Chen%5D(%2Fprofile%3Fid%3D~Ziyi_Chen2)).
- 2 Z. Hu, T. Zheng, V. Viswanathan, **Z. Chen**, R. A. Rossi, Y. Wu, D. Monocha, and H. Huang, “Towards optimal multi-draft speculative decoding,” in *International Conference on Learning Representations (ICLR)*, 2025.  URL: [https://openreview.net/forum?id=9KxnxW0BA5&referrer=%5BAuthor%20Console%5D\(%2Fgroup%3Fid%3DICLR.cc%2F2025%2FConference%2FAuthors%23your-submissions\)](https://openreview.net/forum?id=9KxnxW0BA5&referrer=%5BAuthor%20Console%5D(%2Fgroup%3Fid%3DICLR.cc%2F2025%2FConference%2FAuthors%23your-submissions)).
- 3 **Z. Chen** and H. Huang, “Accelerated policy gradient for s-rectangular robust mdps with large state spaces,” in *International Conference on Machine Learning (ICML)*, 2024.  URL: <https://openreview.net/forum?id=J16WEPdqHJ>.
- 4 **Z. Chen**, Y. Wen, Z. Hu, and H. Huang, “Robust reinforcement learning with general utility,” in *Advances in Neural Information Processing Systems (Neurips)*, 2024.  URL: [https://openreview.net/forum?id=8Uyfr5TcNR&referrer=%5BAuthor%20Console%5D\(%2Fgroup%3Fid%3DNeurIPS.cc%2F2024%2FConference%2FAuthors%23your-submissions\)](https://openreview.net/forum?id=8Uyfr5TcNR&referrer=%5BAuthor%20Console%5D(%2Fgroup%3Fid%3DNeurIPS.cc%2F2024%2FConference%2FAuthors%23your-submissions)).
- 5 **Z. Chen**, Y. Zhou, and H. Huang, “On the hardness of constrained cooperative multi-agent reinforcement learning,” in *International Conference on Learning Representations (ICLR)*, 2024.  URL: <https://openreview.net/forum?id=wFWuX1Fhtj>.
- 6 W. Xian, **Z. Chen**, and H. Huang, “Delving into the convergence of generalized smooth minimax optimization,” in *International Conference on Machine Learning (ICML)*, 2024.  URL: <https://openreview.net/forum?id=QPSEPI9bvp>.
- 7 **Z. Chen**, Y. Zhou, Y. Liang, and Z. Lu, “Generalized-smooth nonconvex optimization is as efficient as smooth nonconvex optimization,” in *International Conference on Machine Learning (ICML)*, 2023, pp. 5396–5427.  URL: <https://proceedings.mlr.press/v202/chen23ar>.
- 8 J. Liu, L. He, Z. Chen, **Z. Chen**, Y. Hao, and D. Jiang, “Context-aware eeg-based perceived stress recognition based on emotion transition paradigm,” in *International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, IEEE, 2023, pp. 1–8.  URL: <https://ieeexplore.ieee.org/document/10388140>.
- 9 Z. Xu, **Z. Chen**, and X. Chen, “Dynamic feature-based newsvendor,” in *ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems*, 2023.  URL: <https://openreview.net/forum?id=SSFshVL6tD>.

- 10 **Z. Chen**, S. Ma, and Y. Zhou, "Accelerated proximal alternating gradient-descent-ascent for nonconvex minimax machine learning," in *IEEE International Symposium on Information Theory (ISIT)*, 2022, pp. 672–677.  URL: <https://ieeexplore.ieee.org/document/9834691>.
- 11 **Z. Chen**, S. Ma, and Y. Zhou, "Finding correlated equilibrium of constrained markov game: A primal-dual approach," in *Advances in Neural Information Processing Systems (Neurips)*, 2022.  URL: <https://openreview.net/forum?id=LZ6AWPfJKkd>.
- 12 **Z. Chen**, S. Ma, and Y. Zhou, "Sample efficient stochastic policy extragradient algorithm for zero-sum markov game," in *International Conference on Learning Representation (ICLR)*, 2022.  URL: <https://openreview.net/forum?id=IvepFxYRDG>.
- 13 **Z. Chen**, Y. Zhou, R.-R. Chen, and S. Zou, "Sample and communication-efficient decentralized actor-critic algorithms with finite-time analysis," in *International Conference on Machine Learning (ICML)*, 2022, pp. 3794–3834.  URL: <https://proceedings.mlr.press/v162/chen22ah.html>.
- 14 S. Ma, **Z. Chen**, Y. Zhou, K. Ji, and Y. Liang, "Data sampling affects the complexity of online sgd over dependent data," in *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2022, pp. 1296–1305.  URL: <https://proceedings.mlr.press/v180/ma22a.html>.
- 15 **Z. Chen**, Y. Zhou, T. Xu, and Y. Liang, "Proximal gradient descent-ascent: Variable convergence under kŁ geometry," in *International Conference on Learning Representation (ICLR)*, 2021.  URL: <https://openreview.net/forum?id=LVoTkZmYyDi>.
- 16 S. Ma, **Z. Chen**, Y. Zhou, and S. Zou, "Greedy-gq with variance reduction: Finite-time analysis and improved complexity," in *International Conference on Learning Representations (ICLR)*, 2021.  URL: https://openreview.net/forum?id=6t_dLShIUYZ.
- 17 C. Chen*, **Z. Chen***, Y. Zhou, and B. Kailkhura, "Fedcluster: Boosting the convergence of federated learning via cluster-cycling," in *2020 IEEE International Conference on Big Data (Big Data)*, * These authors contributed equally to this work., 2020, pp. 5017–5026.  URL: https://ieeexplore.ieee.org/abstract/document/9377960?casa_token=2806KKk2m1kAAAAA:hygjJAU8WFn1Rgo3Juqi_thMrSLw11fm8GGKXlvC15xpP2320BSVV1dWymdzw8cH6DjbIvuuAy8.

Other Research Experiences

- 2025  Application of reinforcement learning to diabetes control (Ongoing)
- 2023-2025  Mentor Ph.D. students (Publications: 2 **ICML**, 1 **Neurips**, 1 **ICLR**)
- 2021  Contribute to NSF CCSS funding (accepted) on multi-agent reinforcement learning.

Honors and Awards

- 2023  A **Top Reviewer** at The Conference on Uncertainty in Artificial Intelligence (**UAI**)
- 2015  Certificate of Appreciation for Contributions to the Math Explorer's Club, K-12 Education and Outreach, Math Department of Cornell University
- 2014  Founder Scholarship, Peking University
-  Pacemaker to Merit Student, Peking University
- 2013  National Scholarship, Ministry of Education (China)
-  Certificate of Volunteer as a Tutor of Advanced Math, Peking University
- 2012  Honor of "Outstanding Graduate Cadre", Xi'an Jiaotong University
- 2011  Honor of "Excellent Student Cadre", Xi'an Jiaotong University
-  National Scholarship (About 50 in 3400), Ministry of Education (China)
-  The Second Prize of Math Modeling Competition in Xi'an Jiaotong University

Honors and Awards (continued)

- 2010
 - The Samsung Scholarship, Xi'an Jiaotong University
 - Honor of "Excellent Students", Xi'an Jiaotong University
 - The First Prize of the 8th Competition of Advanced Mathematics, Xi'an Jiaotong University
 - The First Prize of Physics Competition, Xi'an Jiaotong University
- 2009
 - Peng Kang Scholarship, Xi'an Jiaotong University
 - Honor of "Excellent Students", Xi'an Jiaotong University

Selected Teaching Experiences

- Spring 2024 & 2025
 - Co-teach *Selected Topics in Information Processing*, at University of Maryland
- Spring 2019
 - Teach recitation sections of *Multivariable Calculus for Engineers*, at Cornell University
- Fall 2017
 - *Statistics in Real Life* (invited talk), at the Ithaca High School
(arouse students' interest using interesting examples of machine learning)
- Spring & Fall 2013
 - Teach recitation sections of *Mathematical Analysis*, at Peking University

Service

Program

- Committee The Association for the Advancement of Artificial Intelligence (AAAI)

Console

Reviewer

- Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Uncertainty in Artificial Intelligence (UAI)
- Transactions on Machine Learning Research (TMLR)
- Artificial Intelligence and Statistics (AISTATS)
- SIAM Journal on Optimization (SIOPT)
- IEEE Conference on Decision and Control
- IEEE Transactions on Signal Processing
- IEEE Transactions on Knowledge and Data Engineering